ActInf Textbook Group ~ Cohort 8 ~ Session 12 (Chapter 6 and overall) ~ 8/21/2025

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Hello. Okay, welcome slashwelcome back to this next phase of activity. Right now, technically, we're in the cohort 8 first meeting back, continuing with basically reviewing the first half of the book and talking about chapter 6, which as I joked a little bit before, it's kind of like an alternative chapter 1. And we can we can talk about why that might be. And just in terms of what is going to be upcoming for the next two months, Andrew Pasha, who's a very wise and and studied facilitator, and I will be alternating. So basically on Thursdays there's a meeting at 15 and 22 UTC and cohort 8 and 9 are in effect and they they switch times but people tend to join at the time zone that works for them rather than the cohort. So, like right now we're in this 15 UTC and then next week at 15 UTC that'll be technically cohort 9 looking at chapter 1. So, feel free to join either or both of the live sessions or you can check back after the live session and and watch the video recording. Um there are a lot of ways to participate in the textbook group. It's more than could fill a day. So figure out how much time and in what ways you want to actively infer this topic. The learning can look like classical undergrad type studying of the textbook, annotating it, all of that, doing things by yourself. however you like to do it, working in your personal knowledge management spaces all the way on through aesthetic expression, uh, writing code. And there's also a lot of collaborative ways that people can contribute to this textbook group and to the open-source active inference ecosystem as well. And that will continue to hopefully get drawn out over the coming weeks. But we're going to be working on code. There's many code examples that are not discussed in the textbook, but we do want to have some of the code examples that exactly recapitulate some of the textbook examples. Uh so in the contribute to the textbook group, there's there's different ideas that that people have listed over the time. And uh one way that we structure the meetings and also just give ways for people who are here or not here to participate is through this questions table. So there's questions for each chapter that people have written. So sometimes they're kind of like just confirmatory like what did what did this mean or what what do they mean by that here? Sometimes they're more advanced like how do these ideas relate to these ideas or how does this equation get reached? Um or just open-ended questions. And so when we're starting a a chapter discussion, it it could tend to look like just, you know, seeing who's there, seeing what's top of mind for anyone who wants to share a reflection that's grounded in the focus topics or or chapter. Uh, and just see what questions and and uh ideas arise right there from whoever wants to begin. And then uh we can go to the questions table and we can either look at the interesting questions that people have upvoted or we could go to new questions or if there's any questions that people who are in this current phase of activity have submitted then this is a good way to kind of like get it down in writing and it's really fun to read through this kind of work and write down okay I don't understand what they mean here and then like that goes from being a known unknown and then you reduce your uncertainty and then now you know you know and that's active inference itself. So it's a fun way to kind of have a little media as the message type experience with the materials. So, all of that being said, there's a ton more to explore in this KOD document that I'm not going to go into right now, but explore around unfold different lists and uh let us just jump right into it. We're talking today about the first half of the book. So chapters 1 through five and also chapter six. So does anyone just want to start with any general t takes or questions or or comments on the structure of the first and second half of the book or on chapter 6 or or I can add a note on that. But is there anything just that that anybody wants to kind of begin with or that that that they want to start with? Yeah, Andreas and then anyone else who raises their hand. >> Yeah, thank you Daniel. Um, yeah, so like I've said, um, I've read the whole book and I've participated in the last, um, you know, half year. Uh, and I wanted to understand the math. Um, and I felt I just didn't understand the math. But I have a PhD in math, so I should be able to understand it, I hope. Uh so uh a few weeks ago I just sat down and I wor realized and there was a very nice video um I I'll have to say who the author was. It just said it much more clearly than the book and then I kind of sorted out. It's like five or six lines uh I could show how I understand it and then maybe you could see if I got it right or not. But it's not in the book. Uh so this is for the variational free energy. So if I could at some point I would like to show that. That sounds good. Let's sort of just stay at least one foot in the book, but continue. >> Yeah. So, this is the foot that's not in the book basically like to see where is this in the book? I can't find it. >> Yeah. The book is not intended to be comprehensive of the whole field. It's it's a kind of like extended length paper that gives many chances to to explore further. So, I mean totally feel free. Well, but so the but the point is is that I read a 200page book. It did not have the five lines of math that I wanted to see, you know. So I would like to, you know, I don't have to show it, but but I think it would save everybody uh it would just be helpful for people to see like, yeah, where is, you know, where's the math, you know, right? So uh that's what I have to offer. I guess that's what I'm interested in. Yeah, great. When when we come to it, the people have have done different walkthroughs of different levels of depth on the free energy um equations. >> I think also maybe I'll just try to hook in with Jonathan. Well, this is from the first part of the book. I think that's the point. I don't know if this is the right day for this, but uh but I know that Jonathan Shock in South Africa, he's interested, I think, to talk about the math. And Octopus had a learning group on the math. I don't know who else would be interested in the math, but that's what I'm interested in. Yeah, great. any any other just any other things people want to just share what what parts they want to focus on overall or or just a general question or or I can kind of make a remark on just the chapter six and chapter one. Yeah, I think what what you're uh hitting at, Andrew, is uh kind of going from from Baze theorem to active inference, right? And there just seems to be this kind of leap where, okay, I think I've got Bay down. I shouldn't say down, but I I'm understanding it enough. And then we get it. It just seems like it gets complicated in a hurry. And then in the textbook there's the the three variations of the equation where it talks about um complexity minus accuracy and energy minus entropy. I I'm guessing that those equations are all saying the same thing but just from a different lens. Is that or a different perspective? Is that right Steph? And then yes we can explore that. All right. Might might be a slightly different topic or point of view here. Um I am interested in getting the getting the intuitive understanding of the connection between thermodynamic entropy and and like a information entropy and where those things are. I don't think equivalent is the right word, but that's the main transformation that I'm I'm wrestling with. >> Great. Yeah, these these are these are all important areas. Andress. >> Yeah, I was on mute. I guess I didn't. Um, uh, with regard to who who spoke be who who who are the two people who spoke just so I get the names right. >> Uh, Mark Roberg. >> Mark. And then uh, >> so Mark I think talked about the math. And it turns out that the book is um it really like it just the real math just starts with one line that uh explains what the free energy is and then you just scoot some terms around and then you get that it's the sum of two um it's the sum of two differences. There's a sensory difference and there's a conceptual difference and you minimize both. So you're minimizing the the discrepancy in the outer world and you're minimizing this discrepancy in your inner world like your model how you know compared to the reality it's supposed to approximate and it's called an approximation which really is not very helpful because it's not an approximation. It's your inner model of what the world is like. So it's the Q let's say you're comparing the Q and the P. And so when you see that math then the whole entropy thing can make sense. That's it's important to see those five lines that I would like to show. Um yeah lot lot to say here. Also it's important to note like free energy variational basian inference isn't introduced in this textbook. There's several decades of work on that topic and active inference is kind of about bringing that into a unifying cognitive cybernetic type framework. But if you have a variational autoenccoder, not not to jump, you know, too technically too quick, but a variational autoenccoder that's doing end to end, whether it's an image coming in and a classification going out or it's sense data coming in and robot action going out, it's already doing active inference. So this is kind of like a quick introductory textbook through that area. But a lot of the work on the more analytical formalism and on the especially the codebased aspects is happening outside the textbook. So let let us not only see it as a detriment that this or that isn't included in the textbook. That's a discrepancy between maybe someone's expectations of a textbook and what the document actually is. But then that's either something that is happening outside of the textbook or it's an opportunity for future contributions like Sanjie's um future book. So I'll just make a general comment on the the structure here and then also about this bay to active inference question. Um so the book has two halves. There's chapter one through five which they talk about in the beginning of the book as kind of being the epistemic. So that's more of like the learning and theory. And then chapters really six through nine are a little bit more practical and model specific. They have a lot of details about what certain kinds of generative models look like and citations to where those can be found in more detail. And chapter 10 is kind of like a summary. So chapter 1 and 10 are like these bookends that are a little bit more big picture. One is introductory. 10 is a little bit more synthetic. And then chapter six is kind of a fun alternate chapter one for for those people who like a little bit more practical orientation or they have experience with systems engineering or other sort of analytic methods because chapter 6 is the recipe for building a generative model. So it's kind of a process of structured thinking of which there are other structured processes as well. So again where somebody sees like this should have been this way or that you know it's like that's your method that that that can coexist chapter 6 gives this method of four steps that in one way or another basically have to come into play to have the active inference generative model. So that we're going to return to that again and again like these questions either have been asked to yield a full generative model or they haven't at which point it's not complete. Um and then about bay and active inference and where does this like free energy value come into play? This is the figure 1.2. two. And this is a motif that structures chapters 2 and three which are on this low road bottomup kind of builders engineer um can't understand what I haven't built existentialist in the sense of like constructivist building things and this and then on the top down from the high road is more of this generic physics of cognitive systems like information geometry and free energy principle and then where they meet is active inference and that's like described in chapter 4. Um how do you get from B theorem to free energy? Well, when you write down some kind of model that's going to do updating, if you're working with enough compute power or a small enough problem to do exact basian calculation, you can just do that. You can just exactly calculate the basian updating. However, there's a wider range of problems where it takes too much resources or it's not plausible to actually compute over certain aspects of B theorem. And so that's where approximate basian computing and the whole probably approximately correct frameworks come into play. And then there are several flavors of heristics and all these different things but one of the key ones is this variational inference again which has existed for decades. It's known as the evidence lower bound and and we can go into more detail but like it just those are some of just putting puzzle pieces out on on the table. Those are some of the things that that come into play. And it isn't like this textbook or even this last five to 15 years of research invented any of these topics. So it's more like a composition of these topics which you can go down different avenues to understand their intrinsic coherence. But then we bring them on the table and we look for their relational coherence in the context of cognitive systems modeling. But cognitive systems modeling including a continuum from something that's like trivially inert to kind of classical and repetitive like a metronome on through more sophisticated cognitive phenomena. all the emotion, attention, anticipation, these kinds of things. So, just trying to to give like many first little pins in. Um, so anyone want to just add add a comment or or go to a Yeah, Steph, go for it. >> This is a question on the uh on the motif, the image. Um um do you typically consider this from the individual perspective or do you also consider this motif from a like a a group of individuals um surprise minimization uh point of view? Like is there scale and variance to bringing this to multiple levels of networks or do you should this typically be viewed as a an individual in infer like um or or do you do you ever do collective many individuals but you look at them as one individual inferring? Yeah, this is a an important question for the collective behavioral studies and for multi-agent systems. So I think a good recent paper again it's this is now 3 years on from the textbook. So there's things that just are outside the scope of the textbook and and happened later but I'll put this one in the chat. This is I think is a a a great one that that talks about that. Um two two ways to explore that. First is through the partition even one is understood to consist of a partitioned number of states. So even if we're talking about one thing we're thinking of that one thing in terms of three even at the most coar grained. Basically it's blanket states the incoming perception like and the outgoing action like and then the internal cognitive like. So even the sort of diatic agent niche is already a four-fold representation with a blanket. So that that's one way to go to say okay well one is already multiple and and that accords with our understanding of you know all systems being composite already and then kind of working up where let's just say you have agent environments you can compose and and sum the values across multiple agents. Now whether that summary statistic is available to the lower level agents that's a design decision you could enact or not and then whether available or only sort of with this sort of with a supervisor like a simulation runner who gets this summary now whether that summary statistic is meaningful or relevant or actionable or better at being used than the estimates of the individuals. Those again are are that's another level of of questions. But the I mean again there's there's so many ways to go with this but the idea would be that the the composability of the graphical models supports kind of zooming in and partitioning a collective down. and then also building more flexibly up. >> Yeah, thank you. >> But this book does not go into any multi-agent aspects at all. It just focuses on on the sort of single organism, but it it it makes some notes. Like if we're looking at um you know a nervous system example like in chapter 5, it's like yeah you you could zoom in and you could you could go make a cellular level model but like again that is the sort of connection between more general hierarchical nested complex systems biology approaches and this is like okay and here's a way where you can compose those models and then that also kind of connects to some more recent work like with the reormalization group and everything. Anyone can can raise their hand or or write a question. These chapter six qu questions are great questions to think about. Um they they set up instead of taking the sort of chapter one literature review approach. This starts to read again more like a set of steps you can follow to do a given modeling process. And whether you only keep this at the level of sketched out on paper or whether you take it on through one of the multiple possible ways to execute this code. Seeing this as recipe creation that gets prepared or not, like there's recipes that you can write out that you can't cook or there's recipes you could cook but you don't have the time or the resources to. It gives it a lot more of a a pragmatic basis in what modelers do. Like active inference isn't just an oracle dis dispensing opinions on different phenomena. It's a modeling framework that modelers use to explore different hypotheses about systems of interest. So the first question which system are we modeling? It's like the generic form of active inference is not making reference to any particular system at all. So there's nothing about any particular system or any aspect of any system that on a first order invalidates anything about the generic formulation. So th this is kind of like I mean again a lot of ways to to to explore this but like starting with a system of interest and then exploring the interfaces that has a lot in common with systems modeling and stock and flow modeling and other ways of doing systems modeling. This is just one that that continues on and connects to a lot of effective software. Um, but yeah, do we want to continue through these questions or is there some other question from the first half of the book or from chapter six that that people would rather look to? Daniel, I I feel like I could ask you you questions and I don't want to be rude uh with the other members here. Um my question in in an act of inference sense would be do you consider there being an end like a goal for yourself relative to the constituents in the room or what is achieved here? And it's kind of more of a personal question like what is what is the textbook review and this type of what what would you prefer to see come out of this? Um, is this a room where one just has the floor to ask questions um where you can bounce around or is it to be that there are so many cohorts going on and it makes sense to be in sequential order to to the book? Um, like what is your preference to get things out of it? because otherwise I would I would start to to to be I could be quite selfish and I don't want to be >> yeah I I'll look to see what other people want to add. Like for me it it gives a time boxed opportunity to go over important material the many many many times which some of us may want to or feel like we need to go over it. like using it to have one foot in the book, rereading an important text and connecting it to more recent and other resources is a valuable exercise. I like to see when in their own motivation and way people who are activated to either show up and learn and participate in open science and and be a part of the community and then also realize that there there's more contributions than not to to create code to do all these things. like week by week people could be bringing lists of questions and short blog posts or research that they're working on along the way and I mean yeah I can I can add more but but this is about as with everything at the institute opportunities for people who want to learn and apply active inference to connect with other people who want to do the same rapidly reduce their uncertainty see about what resources already exist. >> Good. Thank you. Okay. Yeah. She need the >> Yeah, these these are great videos by Shamil Chandaria. I'm amazed the book doesn't include the math. I I hear you. Doesn't start by defining free energy and reorganizing the terms. Well, chapter one is just a background chapter. Chapter two basically does the minimum walk to get to free energy. starting with the base theorem and starting with the exact base and then basically saying well you can't always do exact base. Sometimes you need to make an approximation or you need to have your own internal model of another process. So Andrew, I'm I'm I'm Q is not arbitrary as the book implies. It it's it is selected arbitrarily by the modeler of the agent. That doesn't mean ungrounded or haphazardly per se, but it's it's selected at will by fiat regardless of of what the external world is. Mark Thank you. Um, yeah, thanks for going over these steps here. Um, it I've read the book and you at least that part of the book and several tutorials. Um, I think Smith was the first author, Ryan Smith was the first author on the 2022 tutorial which which I think had helpful elements in it as well. Um it seems like there is a lot that is made of uh the idea that we we can't do exact bays and that we can't have an exact uh I don't I guess it's not the likelihood there there's there's a term that is it's the uh the Q right that is not that we can't have or no P one of the values is not so we we get into this where the variational idea is introduced. But I I think that if I'm right the active inference that I mean that's not necessarily the main point that's a point but the main point is that there is a way to uh minimize that divergence and and that's what the active that's like the the essence of active inference is minimizing that that difference right but I guess the step that I'm confused on is the fact that well there there is a difference between what the true posterior that's what it is right there's a difference between the true posterior and our our approximation of it I I don't know if that's a coherent yeah there's a lot of important pieces there so let's think about some landscape I I always find it interesting that biologists when they're talking about fitness they talk about increasing fitness or like a computer scientist might talk about increasing reward. So that's like hill climbing algorithms. So there's gradients that you can follow to increase uh on that landscape and and to get up to to more rewarding state um or a model that fits better. And then there's this dual way to look at it like flipping that landscape and going downhill. Because if you think about surprise as being basically inversely correlated with model fit, then minimizing surprise is having a better fitting model. If you were minimally surprised by the next observation coming in, you would have the most evidence for the model. um you'd be you know least surprising would be if you perfectly predicted what was about to happen you would have the least possible surprise. Okay. So there's first this core duality between having a model that fits better which you also want to penalize by not overfitting but just leaving that second piece aside for now. You want to have a model that fits better, which is equivalent to being less surprised. Now, which way to go? Do you want to ratchet up how well your model fits or do you want to try to ratchet down how well or how much surprise you're getting from observations? Those are equivalent processes. And what variational inference does is with a distribution that you control Q which is the the approximator distribution. You can make this decomposition where the evidence term like the o how likely overall is this fogyny? How likely overall is it that Bitcoin is exactly this price? Those questions are actually like you could argue whether they're hard or easy, but they're not needed to calculate a gradient with this left term. So that's one of the key things that allows you to if you're working with that surprise bounding approximation to follow a gradient that actually has a very simple structure and can be computed and incrementally optimized without needing to like worry about your absolute elevation so to speak cuz you just have to test. Okay. Should I adjust the knob this way or that way? Not like how loud is it overall? >> So, how is Q under our control? I think that's what you said, right? >> Yeah. >> This is great, by the way. Thank you. Q is the distribution that that the agent is proposing. You know, I don't want to copy it every single time, but we're talking about a map of a real system. So we're not saying like this is what the bird is doing or something. It's like we're modeling the bird this way or we're choosing to build an agent this way or something like that. Um now the world could be handing observations through any number of mechanisms. It could be a neural network. It could be a lava lamp. It could be radioactive decay. It could be running through a list. It could be so the the way that the world generates those observations could be any numbers of way. But by having a distribution that's controlled like I'm going to model those incoming values with a Gaussian distribution and then I'm going to shift the Gaussian move the mean and then I'm going to adjust the standard deviation like make it wider or thinner. So it's like I have two knobs on my Gaussian. That's the one I control. And then I'm going to kind of like do this etch a sketch controlling those two knobs to make the best fitting approximation Gaussian for whatever data are coming in. And again, the data coming in could be like, you know, 333 9999 3 I3399 and you'd get all these secondary like should I should I have a big Gaussian that fits over both or should I do a tight one and then flip it. So there's all these statistics questions that come into play. But the big picture is you have a welltempered etch sketch Q that you know the properties of and control the properties of and got to choose completely and then because the agent is making updates only on Q and action only updating it's it's it's taking the external observation in but not modifying observations like the thumb isn't on the scale but by saying okay well now I'm going to change the world change the mind those are the two options I'm going to have a unified energybased model minimization process where to some extent I'm going to either be updating beliefs on Q you know which but it could be beliefs about action as well but not going to have my thumb on the scale and I'm not going to engage with the possibility of of direct telekinesis. Like only through action will the generative process out there in the world be modified. But I'm going to adjust my sensemaking frames andor my action selection beliefs based upon the the pointlike observations that I'm getting in. that gives this like wellont controlled setting where even if the the process out in the world were were changing how it were being generated still the model of the agent would be kind of like well typed that's a good explanation so I just and I don't want to dominate the conversation here but I just want to follow up here Q includes obviously predictions um and you were talking about precision of those predictions when you were talking about uh the type of distribution right am I going to have a a wide or am I going to have a narrow that that reads as precision is that right and and that precision is also a prediction right so so I'm making these and so you would you see that as separate or you see that as just kind of all inclusive and then finally it seems like the action is also embedded in the model if that's the right way to describe it. So it's so so the model Q has sort of I mean you can kind of bake everything into Q is that right? Yes. Like the agent is the joint distribution of all of those things. It's a joint distribution over perception, cognition and action. And so doing optimization on that whole joint distribution like th that's the variables you need to represent in code to have the agents is its whole joint Those that's the joint distribution that constitute the generative model. And so that could have if that could have just um it could be like a spike like belief. So infinite precision moving just exact beliefs around. Or you could add another parameter to have uh more of a a two parameter belief moving around like the mean and the variance like the Gaussian. Or you could have all of these compositions of different variables. Again, you get into this secondary question of model selection like what is the relative model adequacy or what are the resource costs associated with these different models or is there a fair way to reward them for fitting well but penalizing having more parameters that's like the bay information criterion those are sort of secondary questions arising but whatever joint distribution you have is the generative model for for the agent Andre Yes. Uh I had a question about um equation 2.5 which you showed earlier the three equations and then um >> okay >> you hear me? >> Yeah. 2.5 and then the middle one it's about complexity and accuracy. I'm wondering how to derive that. That's the one I couldn't do. Sure. This was um Yeah. Did Did you look at Well, you looked at Jonathan Shock's material. >> Not yet. I I just want to if you go to the equation. >> Yeah. Okay. >> Here was here was Yakob, I believe, wrote these. Also just just as a sort of note, I'm I'm currently vibe theorem proving with lean and exploring some some more distilled representations. >> Uhhuh. >> So I so you know for to for anyone who who wants >> So this is very good. >> But yeah, >> where where are these notes? Where are they available? >> Um it's in the the canvas for equation 2.5. >> I see. You open these up and then it gives derivations. And so this is there's a there's a page on equations. Is that right? >> Yeah. And the equations page has like the latte code. Um >> Okay, good. >> Yeah. And then it's in here. I I I know it's kind of scrolled away in a little canvas, but not not everything can be on the the main table. And and also again, this is like, you know, 30 to 70 years. These are stable derivations. This isn't intended to be new ground in the book. It's just reviewing that that free energy functional has these different compositions possible and then >> yeah I wanted to find this so I didn't know this. Okay, thank you. That's very helpful. >> Yeah and and there's like there's statistics textbooks we we we could probably link to connect that might go into a little bit more detail but but I found Yakob's uh moves here quite useful. Very good. >> And I think Nick had a question in the chat. I don't know if it was answered already, but >> is Q the same policy as RL? Um, >> no. So there's uh policy. If you're talking about a belief distribution over policies, yes, but Q more generally is just used to refer to any belief distribution. There's Q-learning in machine learning which is um also kind of gets gets close but as with many machine learning things and and reward learning like there's this another reward um distribution layered in that we don't really engage with here directly but if you um look at the Q-learning learning. You see a lot of com uh a lot of uh concordances with with how belief uh beliefs about state transitions work here. Basically, a reward is associated with state-to-state transitions. So you have you know you say okay I'm in this um grid world and the transition from here to here's my reward value on that in in the approach that's taken like in the POMDP in the textbook the B matrix is just the state transition matrix so that's just beliefs about policy conditioned beliefs about state transitions then expected states lead to expected outcomes which then get evaluated in terms of pragmatic value and how well those expected outcomes would align with preferences. So that's how there's this kind of Q-learning like dynamic but again it's not the variational Q but it doesn't involve this trained reward function. It just uses latent state transition matrix. What observations do I expect those states to transition to to admit? How would I feel about those observations if I were to get them instead of if I take this action, what states do I expect it to go to? And what reward do I assign or receive from that transition? How do you reward a machine? Keep it running. Update software Lee. Yeah. Oh, wait. Unmute and then go for it. >> Okay. Sorry. So, I have a questions about the Mac blanket. >>

Yes. So like in on in the first step we have to define the system right as uh like um one of the way to define system is to um yeah one of the way to define the system is to focus or to define the mark of blanket. It makes me think about like a language model. We can consider it as an entity living in a text word which it can only access to a text and it can only take action as generating text. So can we consider the text itself the mark of blanket or is the environment or is both? >> Yeah, good question. So this this uh just about the LLM specifically this this >> paper kind of starts to explore that and there's been later exploration. Um there's kind of two ways to address this question of Markov blankets when it comes to real systems. The first one is like interface design and just thinking empirically what are the inputs and the outputs. So in that sense just what are the what is the model receiving your prompt what is it outputting that language. So you just think about this as like API calls. So just do a first approximation. That's the blanket. And then also there's blanket discovery methods that you can use to explore like what in a little bit of a semi or unsupervised way. What are underlying what are the underlying things that seem to be sending messages? But when it comes to to to modeling pretty much just think what observations are coming in for that agent what is coming out and if you have fully enumerated that you describe the Marov blanket. >> Yeah. Then well I wonder then the because um like I think uh my intuition is just that a language model should have a generative model of a language environment. But if we consider the language that it get as input and output as it blanket then it cannot access to the real environment that is uh um human user human customer. Is that right? And it can only model the real environments as in the real human interactions by um um processing the information it gets from the text. >> Well, it's like I'm only getting your audio and visual and I'm only sending you audio and visual. >> Yeah. So similarly an LLM might only receive audio or just text. >> yes, but like um as a an active inference agents, one should uh one should model the environments outside of its direct perception. Right? And for the language model for the LRM it means it needs to simulate the environment outside the text. Is that right? Just like I want to simulate like information from you. I I can only access your image, your videos, your audio, but I I want to maybe think about your thought. >> Yeah, there's there's a lot of ways to go. I'd say if evolution were acting on a given agent to exist within a niche, then there starts to be an aboutness of its predictions about the world outside. But if it's just whimsically placed in an environment, its predictions might be totally irrelevant. So it doesn't need to be doing it like again the express the the space of models you can express includes the nonsensical ones. Like you could have an LLM that's trained on language one getting input from language 2. So like it's the internal model that it is a map of maps could map very well or not well to another situation but maybe sketch it out or or make some open- source code or write it out a little bit more fully and and we can explore it like there's so many things that it's just hard to convey through real time but the beauty is that we can diagram it and actually code it. >> Yeah. I I somewhat understand what you mean. Yeah. Thank you. >> Cool. Okay. So, um yeah, I I hope people can use the the the rhythm of what we do here to learn on their own trajectories. If it just the more preparation and work up front that people do like the more fun it is. Coming with written questions or coming with something that you worked through to share is awesome. So I hope that that people scan over what is already existing. if you want, you know, if you don't have access, request access to to make edits, right? Write questions and and um I'm looking forward to where we go and and and if you show up later time, you'll enjoy Andrew's facilitation. Otherwise, just look at the meeting page and kind of please look at what we're going to be at least trying to connect with and have some quotes or some specific pieces that we can go to. But certainly, as it's now been like three and a half years since the textbook, we increasingly draw on other resources outside of the book. And uh there are other learning and and coding and development experiences at the institute and elsewhere. So if somebody just like wants to be that zero to one and start up a math group or start up another project or work on some other focus like that's all great. Not everything has to happen within the textbook group. Not everything has to happen within the institute. Awesome. Yeah, Steph, go for it. >> Daniel, question. Um, I want to start a group to help to do let's see what's where's the the best place to start a group or to like demonstrate interest in a particular direction. Yeah, I'd say go to welcome.active inference.institute. That'll kind of just take you to this living document to our main page. And then project preparation. If you want to submit a project and have it visible to others, then just completing this form will will get you on and then that and then people can um it it basically will be included in the activities and then the activities page has all of these projects. >> Cool. All right. Thank you. See you all. Thanks for the chat. Bye. Thanks so much.